CambridgeMATHS
Practice HSC Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams in pencil
- NESA-approved calculators may be used
- The HSC Reference Sheet may be used
- In Section II, show relevant mathematical reasoning and/or calculations

Total marks – 100

- Section I is worth 10 marks
- Section II is worth 90 marks

Section I

Attempt Questions 1–10
Allow about 15 minutes for this section

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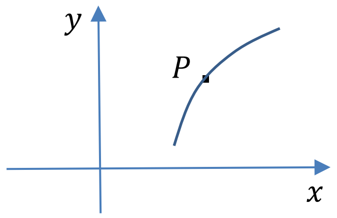
1. What will an investment of $\$50000$ be worth in 5 years if it is invested at 6% $\mathrm{p.a.}$ with interest compounded monthly?

A. $\$51262.56$
B. $\$66911.28$
C. $\$67442.51$
D. $\$71589.42$

2. Which statement is true at the point P on the curve $y = f(x)$?



A. y' and y'' are both positive

B. y' and y'' are both negative

C. y' is positive and y'' is negative

D. y' is negative and y'' is positive

3. The results of an English examination were normally distributed. The mean and standard deviation were 63 and 16 respectively. If Peter's z-score was -1.5, what was his actual examination mark?

A. 39
B. 42
C. 47
D. 57

4. The line ℓ has equation $x + 2y + 3 = 0$. If m is the gradient of ℓ and θ is the *acute* angle between ℓ and the x-axis, then which statement is true?

A. $m > 0$ and $\theta > 45°$

B. $m > 0$ and $\theta < 45°$

C. $m < 0$ and $\theta > 45°$

D. $m < 0$ and $\theta < 45°$

5. If $F(x)$ is a primitive of $f(x)$, which statement is true?

A. $\int_a^b F(x)\, dx = f(b) - f(a)$

B. $\int_a^b f(x)\, dx = F(b) - F(a)$

C. $\int_a^b F(x)\, dx = f(a) - f(b)$

D. $\int_a^b f(x)\, dx = F(a) - F(b)$

6. Which expression is equal to $\sin x \sec x$?

A. $\operatorname{cosec} x$

B. $\cos x$

C. $\tan x$

D. $\cot x$

7. Consider the point $P\big(a, f(a)\big)$ on the curve $y = f(x)$. If $f'(a) = 0$ and $f'(x)$ is positive on both sides of P, which statement best describes P?

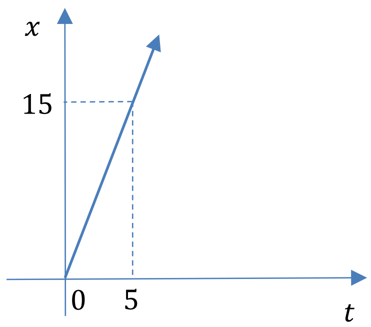
A. P is a point of inflection

B. P is a horizontal point of inflection

C. P is a minimum turning point

D. P is a maximum turning point

8. The displacement-time graph of a particle moving in a straight line is shown below. The displacement is measured in metres and the time elapsed is in seconds.



What is the velocity of the particle?

A. $20\ \mathrm{ms}^{-1}$

B. $10\ \mathrm{ms}^{-1}$

C. $45\ \mathrm{ms}^{-1}$

D. $3\ \mathrm{ms}^{-1}$

9. What is the gradient of the tangent to the curve $y = (x-2)^3$ at its y-intercept?

A. -8

B. -6

C. 12

D. 300

10. Which function is a primitive of xe^{x^2}?

A. $\frac{1}{2}e^{x^2}$

B. $\frac{1}{2}xe^{x^2}$

C. $(x^2-1)e^{x^2-1}$

D. $\frac{e^{x^2+1}}{x^2+1}$

Section II

Attempt Questions 11–31
Allow about 2 hours and 45 minutes for this section

Question 11 (4 marks)

The heights, in cm, of a squad of 10 basketballers are listed below in ascending order:

$$178, 180, 183, 183, 184, 185, 187, 187, 190, 198$$

(a) What is the median height? **1**

(b) Find Q_1, Q_3 and hence the interquartile range. **1**

(c) Draw a box plot of the given data. **1**

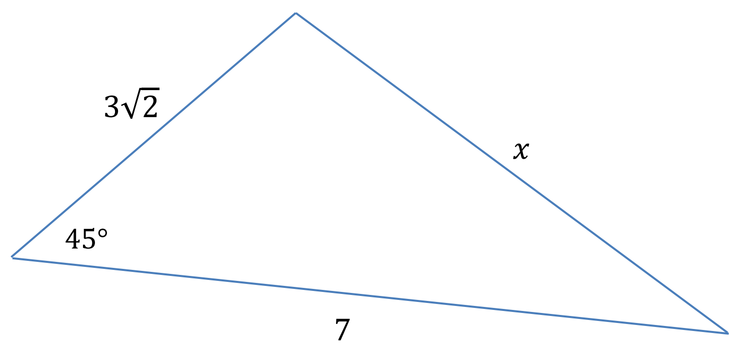
(d) Determine whether there are any outliers amongst the heights using the criterion that an outlier is a score below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$. **1**

Question 12 (2 marks)

Lucas has opened an investment account. He will invest $\$A$ at the end of each month for 4 years at an interest rate of 0.5% per month, with interest compounded monthly. He wants the future value of the account to be $250 000. If the future value of $1 is $54.0978, find the value of A correct to two decimal places. **2**

Question 13 (3 marks)

Use the cosine rule to find the value of x in the triangle below. **3**



Question 14 (3 marks)

Consider the triangle ABC with $AB = 20, BC = 15$ and $\angle BAC = 30°$. **3**

Find, correct to the nearest degree, the two possible sizes of $\angle ACB$.

Question 15 (4 marks)

(a) Solve the equation $|x - 2| = 2$. **1**

(b) Sketch $y = |x - 2|$ and $y = 2$ on the same number plane, clearly indicating the coordinates of the points of intersection. **2**

(c) Hence solve $|x - 2| \geq 2$. **1**

Question 16 (3 marks)

A new car depreciates in value by 13.5% p.a. How long will it take for the car's value to fall below 20% of its original value? **3**

Write your answer as a whole number of years.

Question 17 (3 marks)

The table below shows the probability distribution of a discrete random variable X.

x	0	2	5	9
$P(X = x)$	0.3	k	$2k$	0.1

(a) Determine the value of k. **1**

(b) Calculate $\mathrm{E}(X)$. **2**

Question 18 (4 marks)

Find:

(a) $\frac{d}{dx}(\ln 5x)$ **1**

(b) $\frac{d}{dx}(\cos 5x)$ **1**

(c) $\frac{d}{dx}((5x)^2)$ **1**

(d) $\frac{d}{dx}\left(\frac{1}{x}\right)$ **1**

Question 19 (4 marks)

Find:

(a) $\int \frac{2}{\sqrt{x}}\,dx$ **2**

(b) $\int \frac{1}{7x+5}\,dx$ **2**

Question 20 (4 marks)

Consider the function $y = 2 + 2\sin 4x$.

(a) Sketch the graph of the function for $0 \leq x \leq \pi$. **2**

(b) Use your sketch to solve the equation $\sin 4x = 0$ for $0 \leq x \leq \pi$. **2**

Question 21 (3 marks)

It is found that the waiting times for patients to see a doctor at a medical centre are normally distributed with a mean of 17.6 minutes and a standard deviation of 5.2 minutes. Find, correct to 4 decimal places, the probability that a patient will wait more than 15 minutes. **3**

Question 22 (4 marks)

Consider the parabola $\mathcal{P}$ with equation $y = 2x - x^2$ and the line ℓ with equation $y = -x$.

(a) By solving simultaneously, find the points of intersection of $\mathcal{P}$ and ℓ. **1**

(b) Find the area of the region enclosed by $\mathcal{P}$ and ℓ. **3**

Question 23 (6 marks)

Suppose we have an arithmetic series with $T_6 = -10$ and $T_{21} = 50$, and an infinite geometric series with $T_1 = 32$ and common ratio $\frac{3}{4}$.

(a) Find the limiting sum of the geometric series. **1**

(b) Show that the sum of the first n terms of the arithmetic series is $S_n = 2n^2 - 32n$. **3**

(c) Find the value of n for which the sum of the arithmetic series is three times the limiting sum of the geometric series. **2**

Questions 24 (5 marks)

In a bag there are 4 red marbles, 3 green marbles and 3 yellow marbles. Three marbles are chosen at random without replacement. Determine the probability that:

(a) all three marbles are yellow **1**

(b) at least one red marble is chosen **2**

(c) one marble of each colour is chosen. **2**

Question 25 (6 marks)

The random variable X has probability density function

$$f(x) = \begin{cases} \frac{1}{20}x & \text{for } 3 \leq x \leq 7 \\ 0 & \text{otherwise.} \end{cases}$$

Find, correct to 2 decimal places:

(a) $P(3.6 \leq X \leq 5.6)$ **2**

(b) $\mathrm{E}(X)$ **2**

(c) $\mathrm{Var}(X)$ **2**

Question 26 (3 marks)

Of all the First-Year students at one of Sydney's major universities, 80% are local students of whom 10% hold a scholarship. Of the non-local First-Year students, 75% are scholarship holders. If a randomly chosen student holds a scholarship, what is the probability that he or she is a non-local student? **3**

Question 27 (3 marks)

Suppose that A and B are independent events. **3**

Given that $P(A) = 2 \times P(B)$ and that $P(A \cup B) = 0.52$, find $P(A)$.

Question 28 (4 marks)

Water is being pumped from a reservoir. The volume V megalitres of water in the reservoir after t hours is decreasing at the rate of $(360 - 30t)$ ML/hour.

(a) Find T, the number of hours for which the pump is operating. **1**

(b) Given that the volume of water remaining in the reservoir after the pump stops is 2400 ML, derive a formula for the volume V of water in the reservoir after t hours of pumping, where $0 \leq t \leq T$. **3**

Question 29 (6 marks)

Spiro is a property developer. He is going to borrow \$5 million from the Acme State Bank at an interest rate of 3.6% pa. He intends to make an annual repayment of $\$Y$ at the end of each year for 10 years. Interest is calculated at the beginning of each year based on the amount still owing and then added to the account.

Let $\$A_n$ be the amount owing after the nth repayment.

(a) Write down expressions for A_1 and A_2, and hence show that
$A_3 = 5 \times 10^6 \times 1.036^3 - Y(1 + 1.036 + 1.036^2)$. **2**

(b) Hence find the annual repayment, correct to the nearest dollar. **3**

(c) Calculate, correct to the nearest dollar, the amount owing after 5 years. **1**

Question 30 (10 marks)

Consider the function $f(x)$ defined by $f(x) = 8xe^{-0.4x}$.

(a)	Show that $f'(x) = (8 - 3.2x)e^{-0.4x}$.	**1**
(b)	Hence find the coordinates of the stationary point and determine its nature.	**3**
(c)	Over what interval is $f(x)$ is increasing?	**1**
(d)	Sketch the graph of $y = f(x)$.	**2**
(e)	Over what interval is the rate of change of $f(x)$ increasing?	**3**

Question 31 (6 marks)

A piece of wire of length 24 cm is to be cut into two pieces. One is to be bent to form a square of side x cm and the other an equilateral triangle of side y cm. Let $A_1(x)$ and $A_2(y)$ be the respective areas of the square and equilateral triangle.

(a)	Show that $A_2(y) = \frac{\sqrt{3}}{4}y^2$.	**1**
(b)	Show that the sum S of the two areas is given by $S = x^2 + \frac{4\sqrt{3}}{9}(6 - x)^2$.	**2**
(c)	Hence show that S is at its minimum when $x = \frac{8\sqrt{3}}{11}\left(9 - 4\sqrt{3}\right)$.	**3**

End of examination